

Precision Landing for Unmanned Aerial Vehicles

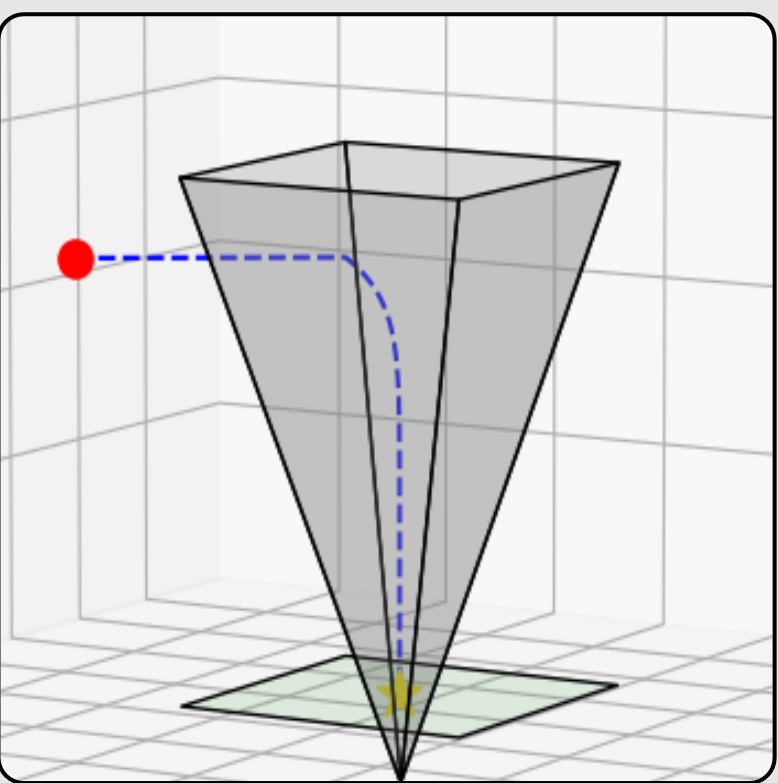
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Autonomous precision landing is needed to improve overall system and mission reliability for Unmanned Aerial Vehicles (UAVs). This method combines a dynamic constraint with computer vision capabilities to shape a vehicle's trajectory in real-time for a safe and accurate descent during autonomous flight. Initial test flights indicate improvement in both precision and vision reliability. The development strategy allows for integration onto current UAV swarm and sensing missions and opportunity for further advancements.

INTRODUCTION

This project includes the development and testing of autonomous precision landing for use in UAV swarm and sensing missions. In house development streamlines integration of landing into existing and future LLNL platforms.

METHODS



Dynamic Constraint

The system is regulated using closed loop control bound by a pyramid constraint to shape its trajectory in real-time over the target.

Figure 1. Simulated trajectory shaping using a pyramid constraint.

Optimization

The constraint ensures that the drone descends from above the target to increase accuracy.

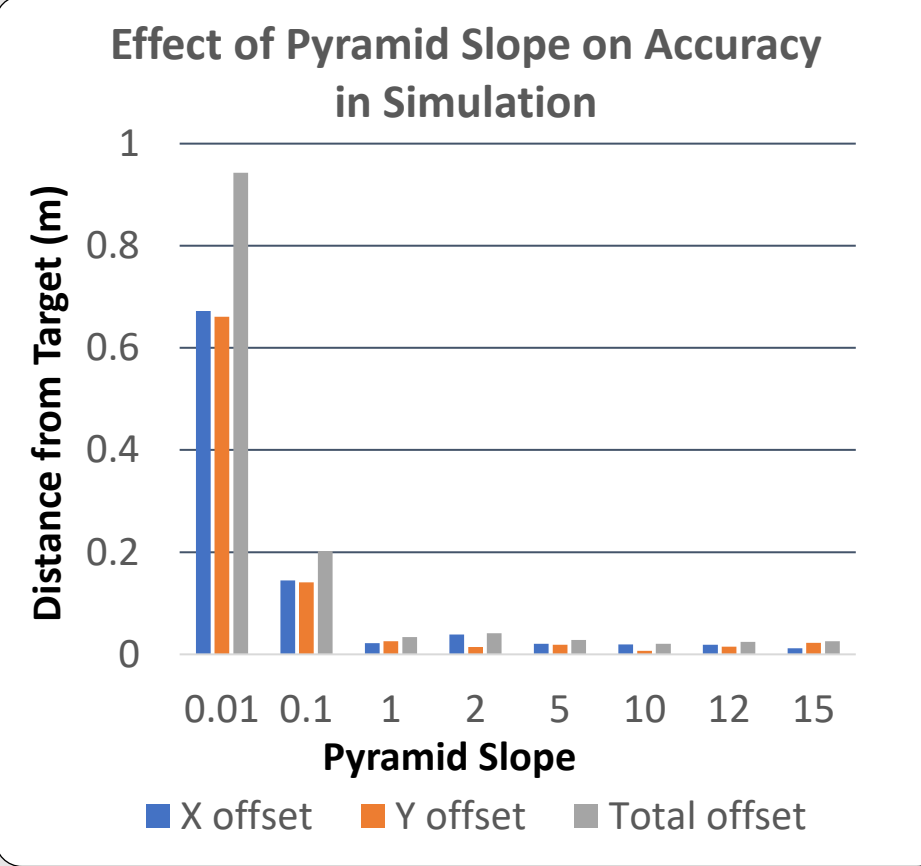
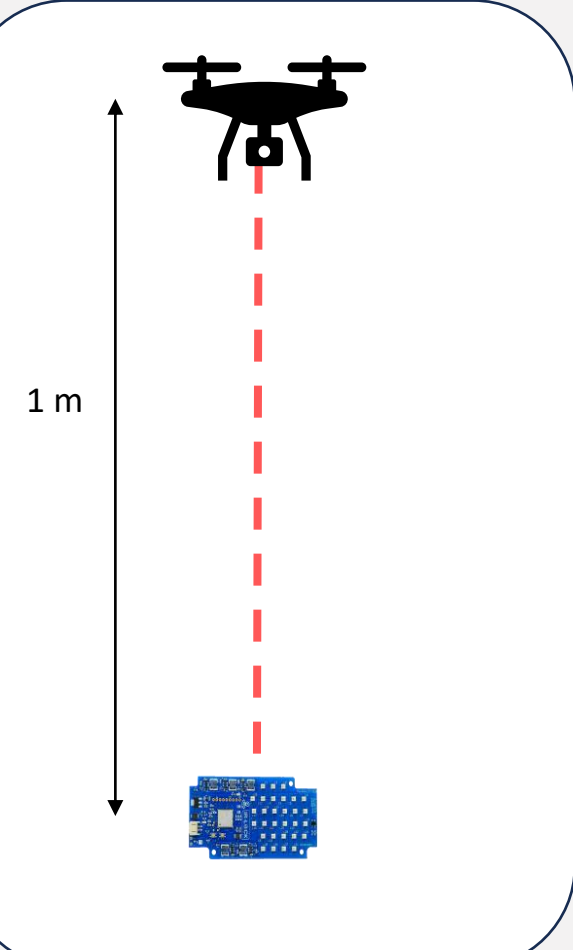


Figure 2. Optimization of pyramid slope.



Computer Vision

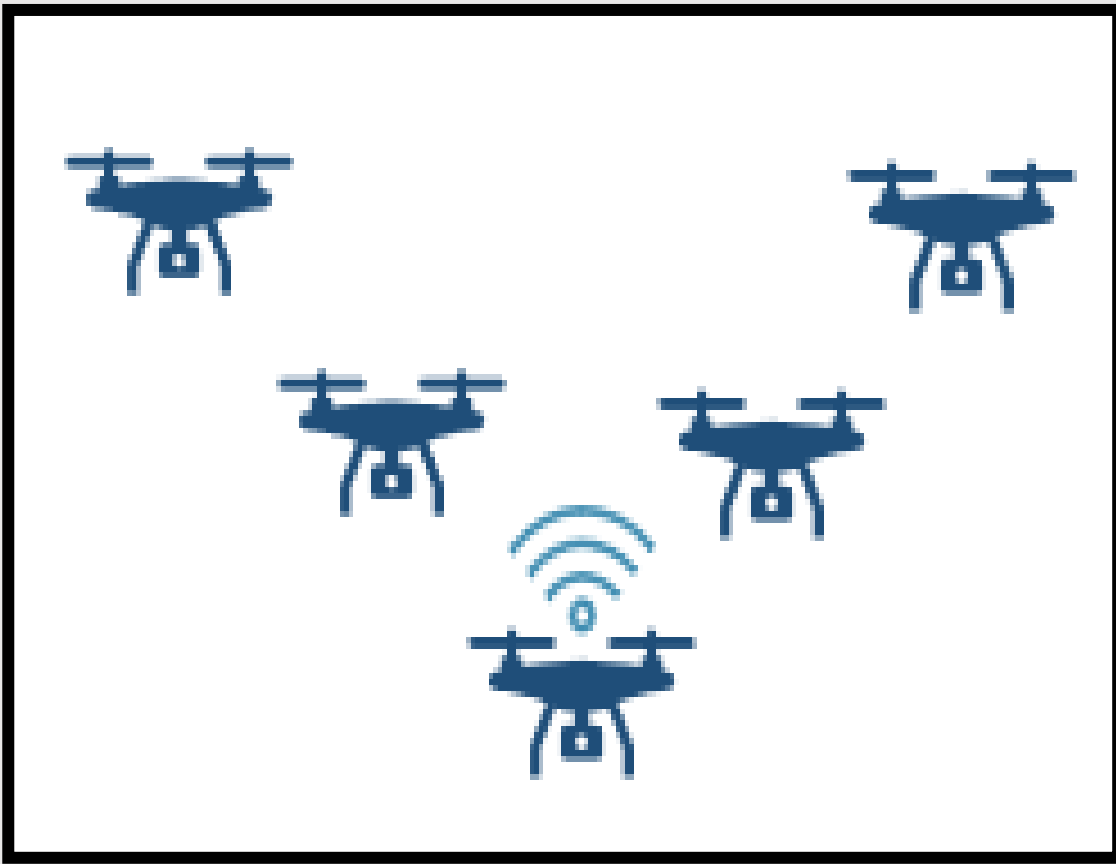
Within a meter of the landing position, the drone uses computer vision to detect an IR beacon for final precision.

Figure 3. IR beacon sensing setup.

PURPOSE

UAV Swarm

Increased landing precision in UAV swarms decreases the likelihood of collisions. Swarm missions will also be more reliable and recoverable with precise landing capabilities.



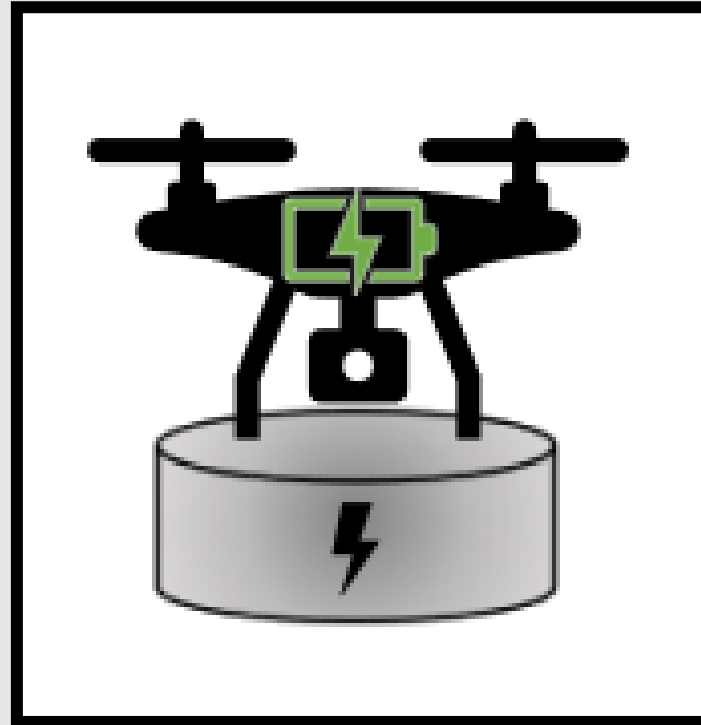
Sensing Missions

UAVs are used in critical missions using ground penetrating radar (GPR) systems developed by LLNL.

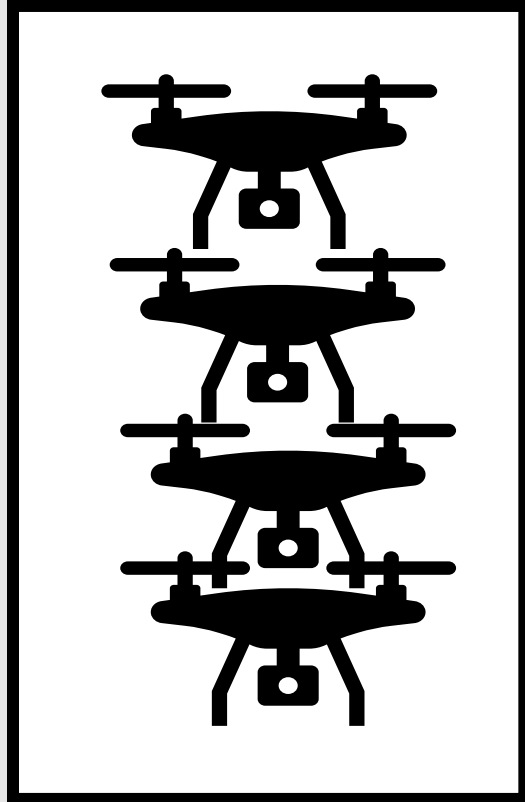
Autonomous precision landing increases the usability of the platform by removing the necessity of human intervention.

The new landing protocol will also increase autonomy with collaboration between air and ground vehicles (Fig. 7).

Future Work



Wireless charging



Efficient storage and deployment

RESULTS

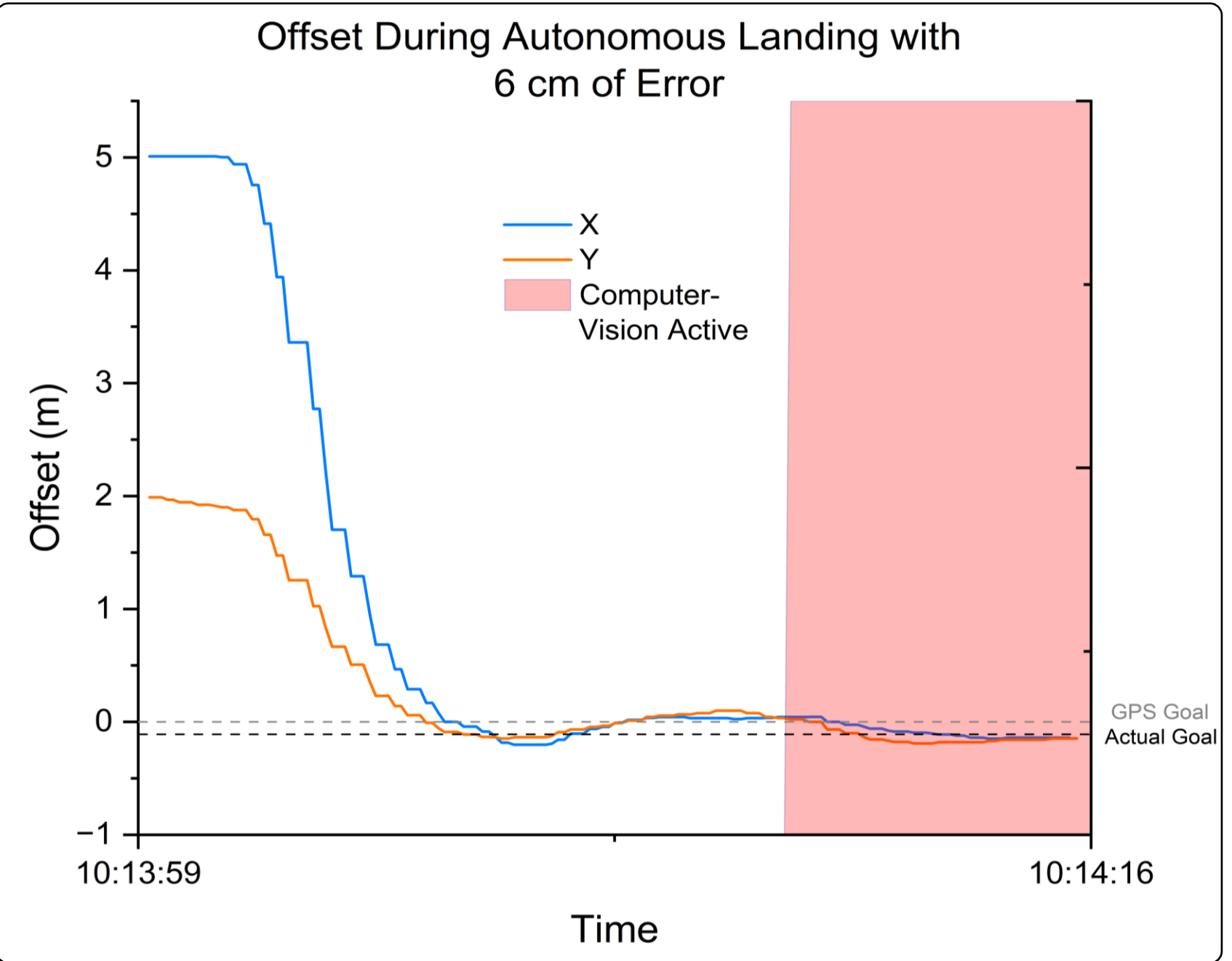
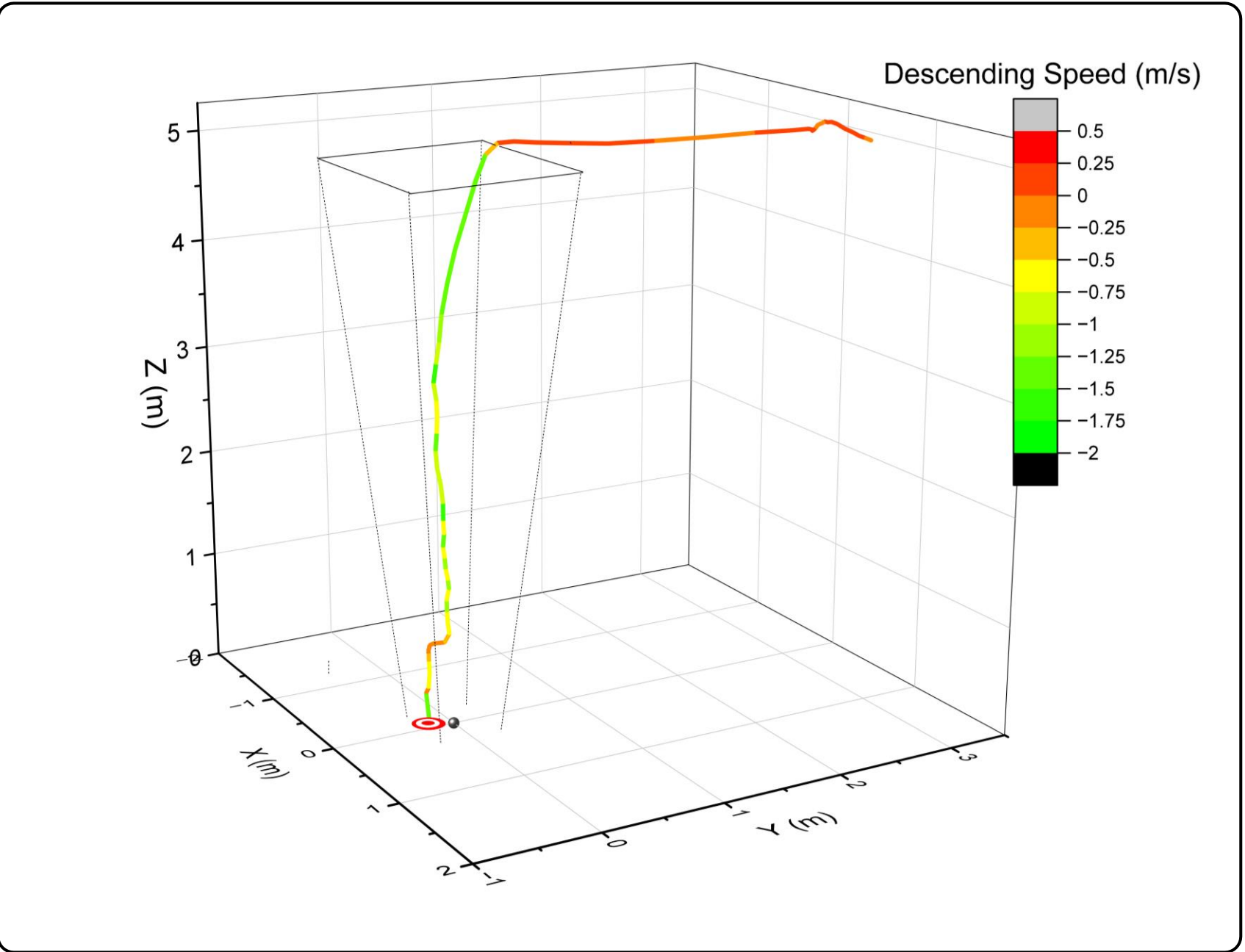
SUMMARY

- Data from 11 successful landing tests in the OS-150 Robotics Laboratory (Drone Pen) and the North Buffer Zone (NBZ).
- Showed improvement from baseline testing in landing precision and computer vision reliability.
- Tests confirmed successful integration into current LLNL swarm software.

Table 1. Compiled data from 11 test flights and comparison to previous solution.

Average Position Error	10 cm	↓ 5-10 cm
Average Heading Error	2.6°	No Data
Beacon Detected	100%	↑ 50%

Drone Trajectory During Landing Test



Landing Test on Polaris Ground Vehicle in NBZ



Figure 7. Precision landing test with the Polaris ground vehicle as the landing target.

Figure 6a. (top) 3D Trajectory of one drone during a precision landing in the drone pen. Pyramid constraint is shown surrounding the GPS landing goal.

Figure 6b. (bottom) 2D convergence to the GPS goal and the actual goal during a precision landing test in the drone pen.

CONCLUSIONS

The proposed precision landing algorithm demonstrated improvement of the landing precision and beacon detection for UAVs in autonomous flight. This protocol was developed to allow ease of integration into existing and future LLNL platforms, extending current capabilities and creating space for advancements.

NEXT STEPS

- Further testing to determine ideal IR beacon positioning for accurate landing.
- Integration of mechanical devices such as a landing funnel to enhance precision in critical cases.
- Extending the precision landing to multiple vehicles in swarm flights.

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